

MH2500 Probability and Introduction to Statistics

Slides for Week 8-2

Topic: Joint Distributions

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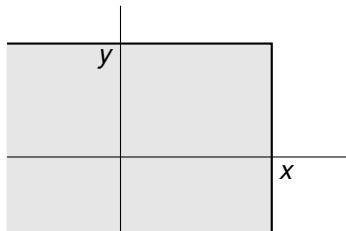
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Joint distribution

Let X and Y be random variables (could be either discrete or continuous). The **joint cumulative distribution function** is defined as

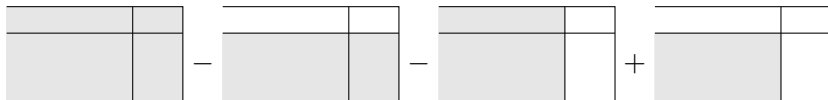
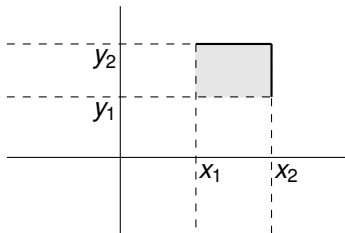
$$F(x, y) = P\{X \leq x, Y \leq y\}.$$

The CDF gives the probability that the point (X, Y) belongs to a semi-infinite rectangle in the plane.



The probability that (X, Y) belongs to a given rectangle is

$$\begin{aligned} P(x_1 < X \leq x_2, y_1 < Y \leq y_2) \\ = F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1). \end{aligned}$$



Connection with Marginal CDF

Given the joint CDF $F(x, y)$ of the joint distribution (X, Y) , how do we find the CDF of X and Y ?

$$\begin{aligned}F_X(x) &= P\{X \leq x\} \\&= P\{X \leq x, Y < \infty\} \\&= \lim_{y \rightarrow \infty} P\{X \leq x, Y \leq y\} \\&= \lim_{y \rightarrow \infty} F(x, y) \\&= F(x, \infty).\end{aligned}$$

Similarly,

$$F_Y(y) = P\{Y \leq y\} = P\{X < \infty, Y \leq y\} = F(\infty, y).$$

Discrete Random Variables

Given X and Y are discrete random variables defined on the same sample space and that they take on values $x_1, x_2, \dots$, and $y_1, y_2, \dots$, respectively, their **joint frequency function** or **joint probability mass function** $p(x, y)$ is

$$p(x_i, y_j) = P\{X = x_i, Y = y_j\}.$$

The probability mass function of X can be obtained from $p(x, y)$ by

$$p_X(x) = P\{X = x\} = \sum_j P\{X = x, Y = y_j\} = \sum_j p(x, y_j).$$

Similarly,

$$p_Y(y) = P\{Y = y\} = \sum_i P\{X = x_i, Y = y\} = \sum_i p(x_i, y).$$

Example

A bag contains 2 white balls and 4 black balls. Draw 2 balls without replacement. Let X be the indicator variable that the first ball is white and Y be the indicator variable that the second ball is white.

The joint frequency function of X and Y is given in the table below.

$x \backslash y$	0	1	$p_X(x)$
0	$\frac{2}{3} \cdot \frac{3}{5}$	$\frac{2}{3} \cdot \frac{2}{5}$	$\frac{2}{3}$
1	$\frac{1}{3} \cdot \frac{4}{5}$	$\frac{1}{3} \cdot \frac{1}{5}$	$\frac{1}{3}$
$p_Y(y)$	$\frac{2}{3}$	$\frac{1}{3}$	

- p_X and p_Y are often referred to as the **marginal** probability mass functions.

Generalization

- The joint CDF for n random variables $X_1, \dots, X_n$ is

$$F(x_1, \dots, x_n) = P\{X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n\}.$$

- The joint frequency function for n discrete random variables $X_1, \dots, X_n$ is

$$p(x_1, \dots, x_n) = P\{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\}.$$

Independent Random Variables

Definition

Say that two random variables X and Y are **independent** if for any two sets of real numbers A and B ,

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}.$$

In other words, X and Y are independent if, for all A and B , the event $\{X \in A\}$ and the event $\{Y \in B\}$ are independent.

Equivalently, X and Y are independent if and only if for any real numbers a, b ,

$$P\{X \leq a, Y \leq b\} = P\{X \leq a\}P\{Y \leq b\},$$

i.e.,

$$F(a, b) = F_X(a)F_Y(b).$$

Independent Random Variables – Discrete Case

Proposition

Two discrete random variables X and Y are independent if and only if for any real numbers x, y ,

$$p(x, y) = p_X(x)p_Y(y),$$

where $p(x, y)$ is the joint probability mass function of (X, Y) .

$y \backslash x$	0	1	$p_X(x)$
0	$\frac{2}{3} \cdot \frac{3}{5}$	$\frac{2}{3} \cdot \frac{2}{5}$	$\frac{2}{3}$
1	$\frac{1}{3} \cdot \frac{4}{5}$	$\frac{1}{3} \cdot \frac{1}{5}$	$\frac{1}{3}$
$p_Y(y)$	$\frac{2}{3}$	$\frac{1}{3}$	

X and Y are not independent.

Example

Suppose that $n + m$ independent trials having a common probability of success p are performed. Let X be the number of successes in the first n trials, Y the number of successes in the final m trials and Z the total number of successes.

- Are X and Y independent?
- Are X and Z independent?

- X and Y are independent:

$$\begin{aligned}P\{X = x, Y = y\} &= \binom{n}{x} \binom{m}{y} p^{x+y} (1-p)^{n+m-x-y} \\&= \binom{n}{x} p^x (1-p)^{n-x} \binom{m}{y} p^y (1-p)^{m-y} = P\{X = x\} P\{Y = y\}.\end{aligned}$$

- X and Z are not independent: It always holds that $Z \geq X$.

Sum of Independent Random Variables

Suppose that X, Y are independent **discrete** random variables.

Then $Z = X + Y$ is a discrete random variable with

$$p_Z(z) = \sum_x p_X(x)p_Y(z-x).$$

Proof:

$$\begin{aligned} P\{X + Y = z\} &= \sum_x P\{X = x, Y = z - x\} \\ &= \sum_x P\{X = x\}P\{Y = z - x\} \\ &= \sum_x p_X(x)p_Y(z - x). \end{aligned}$$

Sum of Independent Poissons

Suppose that $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent. Then $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Proof: Let $Z = X + Y$.

$$\begin{aligned} p_Z(k) &= P\{Z = k\} = P\{X + Y = k\} \\ &= \sum_{i=0}^k p_X(i) p_Y(k-i) = \sum_{i=0}^k \left(e^{-\lambda_1} \frac{\lambda_1^i}{i!} \right) \left(e^{-\lambda_2} \frac{\lambda_2^{k-i}}{(k-i)!} \right) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{i=0}^k k! \frac{\lambda_1^i}{i!} \frac{\lambda_2^{k-i}}{(k-i)!} \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{i=0}^k \binom{k}{i} \lambda_1^i \lambda_2^{k-i} \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k \end{aligned}$$

Conditional Distribution - Discrete

Suppose X and Y are jointly distributed discrete random variables.

Then the conditional probability that $X = x_i$ given that $Y = y_j$ is,

$$P\{X = x_i | Y = y_j\} = \frac{P\{X = x_i \text{ and } Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{XY}(x_i, y_j)}{p_Y(y_j)},$$

assuming that $p_Y(y_j) > 0$.

- This probability is denoted as $p_{X|Y}(x_i|y_j)$.
- This function of x is a **frequency function** since it is nonnegative and $\sum_i p_{X|Y}(x_i|y_j) = 1$.
- If X and Y are independent, then $p_{X|Y}(x_i|y_j) = p_X(x_i)$.

Example

Find $P_{X|Y}(0|1)$ and $p_{X|Y}(1|1)$.

$y \backslash x$	0	1	$p_X(x)$
0	$\frac{2}{3} \cdot \frac{3}{5}$	$\frac{2}{3} \cdot \frac{2}{5}$	$\frac{2}{3}$
1	$\frac{1}{3} \cdot \frac{4}{5}$	$\frac{1}{3} \cdot \frac{1}{5}$	$\frac{1}{3}$
$p_Y(y)$	$\frac{2}{3}$	$\frac{1}{3}$	

- $p_{X|Y}(0|1) =$
- $p_{X|Y}(1|1) =$

Example

If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent, calculate the conditional distribution of X , **given that $X + Y = n$** .

Solution:

$$\begin{aligned} P\{X = k | X + Y = n\} &= \frac{P\{X = k, X + Y = n\}}{P\{X + Y = n\}} \\ &= \frac{P\{X = k, Y = n - k\}}{P\{X + Y = n\}} = \frac{P\{X = k\}P\{Y = n - k\}}{P\{X + Y = n\}} \\ &= \frac{\left(\frac{\lambda_1^k}{k!} e^{-\lambda_1}\right) \left(\frac{\lambda_2^{n-k}}{(n-k)!} e^{-\lambda_2}\right)}{\frac{(\lambda_1 + \lambda_2)^n}{n!} e^{-(\lambda_1 + \lambda_2)}} = \frac{n!}{k!(n-k)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k} \\ &= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k} \end{aligned}$$

Therefore, X **conditioned on $X + Y = n$** follows $\text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$.

Total probability:

By the multiplication rule and the law of total probability,

$$p_{XY}(x, y) = p_{X|Y}(x|y)p_Y(y).$$

$$p_X(x) = \sum_y p_{X|Y}(x|y)p_Y(y).$$

Example

Suppose that a particle counter is imperfect and independently detects each incoming particle with probability p .

If the distribution of the number of incoming particle in a unit of time is a Poisson distribution with parameter λ , what is the distribution of the number of counted particles?

Let N denote the true number of particles and X denote the counted number. Then we are given that the conditional distribution of X given $N = n$ is $\text{Bin}(n, p)$.

- This is a typical model.

For example, N = number of traffic accidents in a given time period with each accident being fatal or nonfatal; and X is the number of fatal accidents.

By the law of total probability,

$$\begin{aligned}P\{X = k\} &= \sum_{n=0}^{\infty} P\{N = n\} P\{X = k | N = n\} \\&= \sum_{n=k}^{\infty} \left(\frac{\lambda^n}{n!} e^{-\lambda} \right) \binom{n}{k} p^k (1-p)^{n-k} \\&= \frac{(\lambda p)^k}{k!} e^{-\lambda} \sum_{n=k}^{\infty} \lambda^{n-k} \frac{(1-p)^{n-k}}{(n-k)!} \\&= \frac{(\lambda p)^k}{k!} e^{-\lambda} \sum_{j=0}^{\infty} \lambda^j \frac{(1-p)^j}{j!} \\&= \frac{(\lambda p)^k}{k!} e^{-\lambda} e^{\lambda(1-p)} \\&= \frac{(\lambda p)^k}{k!} e^{-\lambda p}.\end{aligned}$$

Thus $X \sim \text{Poisson}(\lambda p)$.

- Counted particles: $X \sim \text{Poisson}(\lambda p)$
- Uncounted particles: $Y \sim \text{Poisson}(\lambda(1 - p))$

In fact, they are independent. We need to prove that

$$P\{X = i, Y = j\} = P\{X = i\}P\{Y = j\}.$$

$$\begin{aligned}
P\{X = i, Y = j\} &= P\{X = i, X + Y = i + j\} \\
&= P\{X = i | X + Y = i + j\} P\{X + Y = i + j\} \\
&= \frac{(i+j)!}{i!j!} p^i (1-p)^j \cdot \frac{\lambda^{i+j}}{(i+j)!} e^{-\lambda} \\
&= e^{-\lambda} \frac{p^i}{i!} \frac{(1-p)^j}{j!} \lambda^{i+j} \\
&= \left(\frac{(\lambda p)^i}{i!} e^{-\lambda p} \right) \left(\frac{(\lambda(1-p))^j}{j!} e^{-\lambda(1-p)} \right) \\
&= P\{X = i\} P\{Y = j\}.
\end{aligned}$$

Joint density functions with two continuous random variables

Redo everything for the discrete random variables:

- Marginal pdf and CDFs;
- Independent random variables.
- Conditional distributions
- Sum of independent random variables

Joint Continuous Random Variables

Say that X and Y are **jointly continuous** if there exists a function $f(x, y)$, defined on the entire $\mathbb{R}^2$, such that for any “reasonable” two-dimensional set $A \subset \mathbb{R}^2$,

$$P\{(X, Y) \in A\} = \iint_A f(x, y) dx dy.$$

The function $f(x, y)$ is called the **joint probability density function** of X and Y if $f(x, y)$ is nonnegative and $\int_{\mathbb{R}^2} f(x, y) dx dy = 1$.

In particular, if $A = (-\infty, x] \times (-\infty, y]$,

$$F(x, y) = P\{X \leq x, Y \leq y\} = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv.$$

From the **fundamental theorem of multivariable calculus**, it follows that

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y).$$

wherever the partial derivatives are defined.

Suppose that X and Y have a joint pdf $f(x, y)$. The density functions can be obtained as follows.

$$\begin{aligned} P\{X \in A\} &= P\{X \in A, Y \in (-\infty, \infty)\} = \int_A \int_{-\infty}^{\infty} f(x, y) dy dx \\ &= \int_A f_X(x) dx, \end{aligned}$$

where $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ is the marginal density function of X .

Similarly,

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

Example

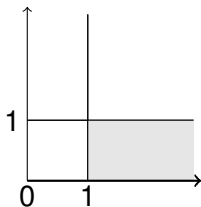
Suppose the joint pdf of $f(x, y)$ is

$$f(x, y) = \begin{cases} 2e^{-x}e^{-2y}, & 0 < x, y < \infty; \\ 0, & \text{otherwise.} \end{cases}$$

Compute (a) $P\{X > 1, Y < 1\}$, (b) $P\{X < Y\}$, (c) $P\{X < a\}$.

Solution: (a) We begin with integrating f over the set

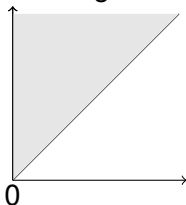
$A := \{(x, y) : x > 1, y < 1\}$.



$$\begin{aligned} P\{X > 1, Y < 1\} &= \int_1^{\infty} \int_0^1 2e^{-x}e^{-2y} dy dx \\ &= \int_1^{\infty} e^{-x} dx \int_0^1 2e^{-2y} dy \\ &= e^{-1}(1 - e^{-2}). \end{aligned}$$

(b) $P\{X < Y\}$

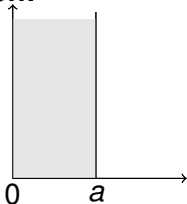
Solution: Integrate f over the set $A := \{(x, y) : x < y\}$.



$$\begin{aligned} P\{X < Y\} &= \int_0^{\infty} \int_x^{\infty} 2e^{-x} e^{-2y} dy dx \\ &= \int_0^{\infty} \left(e^{-x} \int_x^{\infty} 2e^{-2y} dy \right) dx = \int_0^{\infty} \left(e^{-x} \left[-e^{-2y} \right]_x^{\infty} \right) dx \\ &= \int_0^{\infty} \left(e^{-x} e^{-2x} \right) dx = \int_0^{\infty} \left(e^{-3x} \right) dx = \left(-\frac{1}{3} e^{-3x} \right)_0^{\infty} \\ &= \frac{1}{3}. \end{aligned}$$

(c) $P\{X < a\}$

Solution:



For $a > 0$,

$$\begin{aligned} P(X < a) &= \int_0^a \int_0^\infty 2e^{-x} e^{-2y} dy dx = \int_0^a \left(e^{-x} \int_0^\infty 2e^{-2y} dy \right) dx \\ &= \int_0^a \left(e^{-x} \left[-e^{-2y} \right]_0^\infty \right) dx = \int_0^a e^{-x} dx = -e^{-x} \Big|_0^\infty = 1 - e^{-a}. \end{aligned}$$

Therefore,

$$P\{X < a\} = \begin{cases} 1 - e^{-a} & \text{if } a > 0; \\ 0 & \text{otherwise.} \end{cases}$$

Example

The joint density of X and Y is given by

$$f(x, y) = \begin{cases} e^{-(x+y)}, & 0 < x, y < \infty \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability density function of the random variable $Z = X/Y$.

Solution: We find the CDF first and then differentiate it to get the pdf.

For $z > 0$,

$$\begin{aligned} P\{Z \leq z\} &= P\{X/Y \leq z\} = P\left\{Y \geq \frac{X}{z}\right\} = \int_0^\infty \int_{x/z}^\infty e^{-x} e^{-y} dy dx \\ &= \int_0^\infty \left[e^{-x} (-e^{-y})_{x/z}^\infty \right] dx = \int_0^\infty e^{-x} e^{-x/z} dx = \int_0^\infty e^{-x(1+1/z)} dx \\ &= \left(-\frac{1}{1+1/z} e^{-x(1+1/z)} \right)_0^\infty = \frac{1}{1+1/z} = \frac{z}{1+z}. \end{aligned}$$

We find the CDF first and then differentiate it to get the pdf.

- For $z \leq 0$, $f_Z(z) = 0$.

- For $z > 0$,

$$P\{Z \leq z\} = P\left\{Y \geq \frac{X}{z}\right\} = \frac{z}{1+z} = 1 - \frac{1}{1+z}.$$

Differentiate it, we have:

$$f_Z(z) = \frac{d}{dz}P\{Z < z\} = \frac{d}{dz} \left(1 - \frac{1}{1+z}\right) = \frac{1}{(1+z)^2}.$$